

Problem Statement

Product Dissection for top leading Platforms

Welcome to this case study on dissecting and designing products for top leading platforms. In this case study, you will delve into the intriguing world of schema design for a prominent platform of your choice. Your task is to choose a top leading platform, research its features, and meticulously craft a schema design that encapsulates the essence of its functionality. By focusing on key entities, attributes, and relationships, you will gain invaluable insights into how data architecture drives the platform's effectiveness.

Step 1: Choose a Leading Platform

Select a leading platform of your choice, which could span various domains such as social media, e-commerce, finance, or any other industry. This choice will form the foundation of your exploration into its schema design.

Step 2: Research:

Thoroughly research the platform you have selected. Investigate its core features, functionalities, and user interactions. Identify the top features that define its user experience and contribute significantly to its popularity.

Step 3: Product Dissection and Real World Problems solved by the platform

In this step, you will meticulously analyse the platform's standout features and how they provide innovative solutions to real-world challenges. By identifying key functionalities that resonate with users, you'll unravel how the platform effectively addresses problems and enhances user experiences. This dissection will serve as the foundation for understanding how the schema design aligns with the platform's core objectives.

Step 4: Case Study on the real world problems and approach to solving them

In this pivotal step, you will expand on the real-world challenges uncovered in Step 3 through a comprehensive case study. Delve into specific instances where users encountered difficulties and showcase how the platform's unique features provided effective solutions. By dissecting the approach taken by the platform to overcome these challenges, you'll gain a deeper appreciation for the platform's user-centric design philosophy and how it shapes the schema design.

Step 5: Schema Design Based on Top Features

Based on the features you have identified, craft a schema design that reflects the platform's data structure. Focus on the key entities, attributes, and relationships that underpin the chosen features. Your schema should capture the essence of how the platform organises and utilises its data.

Step 6: Rationale Behind the Design

While creating the schema design, consider the rationale behind the platform's choices. Reflect on why certain entities and relationships were chosen and how they align with the platform's goals. This will help you understand the strategic decisions driving the schema's architecture.

Step 7: Create an ER Diagram

Utilise tools like the Miro platform or similar applications to create an illustrative Entity-Relationship (ER) diagram. This diagram should vividly depict the entities, attributes, and relationships present within your schema design. The ER diagram will serve as a visual representation of your insights.

Step 8: Presentation of Findings

Present your findings in a clear and concise manner. Showcase your understanding of how the schema design impacts the platform's functionality and user experience. Explain how your chosen features are integrated into the schema and how the schema's structure supports the platform's objectives.

Task Details:

1. **Answer Submission:** Your submission should include well-structured solutions for all provided questions related to product schema designs.
2. **Video Creation:** Create an informative and engaging video where you thoroughly explain the Case Study.
3. **Depth and Clarity:** Ensure your solutions are detailed and showcase your understanding of product schema design principles. Similarly, in the video, provide clear explanations that are easy to understand for a wide audience.
4. **Creativity Encouraged:** You are welcome to utilise visuals, diagrams, or creative elements to enhance the clarity and impact of your explanations.

Note:

1. Duplicate this document and proceed to write your solutions and prepare your video.
2. Include the video link in this document before final submission.

Best of luck in completing this project and showcasing your prowess in dissecting and designing product schema for leading platforms! **For reference, we have also conducted a case study on Instagram, which you can find below. This case study will provide you with valuable insights into how schema design plays a pivotal role in shaping the functionality and success of a prominent platform**

Capstone project: Product Dissection

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GitHub :- <https://github.com/Dharmeshgadhiya161/Product-Dissection-for-Zepto-Grocery-Delivery-App>



Product Dissection for Zepto Grocery Delivery App

Company Overview:-

Zepto is an Indian quick-commerce company founded in July 2021 by Aadit Palicha and Kaivalya Vohra, offering ultra-fast grocery delivery within 10 minutes. Headquartered in Bengaluru, it operates over 250 dark stores across ten major Indian cities. The company has grown rapidly, becoming India's first unicorn of 2023 with a valuation of \$1.4 billion and reaching \$5 billion by 2024. Zepto has raised over \$1.9 billion in funding and plans to go public in early 2025. The company has more than 1000 staff. It delivers more than 5000 products.

Product Dissection and Real-World Problems Solved by Zepto:-

Zepto App is an ultra-fast delivery platform that operates with a primary focus on delivering groceries and essentials within 10–15 minutes. This app's user-friendly interface allows customers to browse various product categories, search for specific items, and seamlessly add them to their cart. This app in customers can select from thousands of items, such as farm-fresh fruits and vegetables, groceries, dairy, personal care and beauty and wellness products, home essentials, and baby care. This app real-time order tracking like customers can track the status of their orders in real-time through the app. From order confirmation to preparation and delivery, the app keeps users informed every step of the way. This app in the Payment Gateway System allows users to pay for their orders using credit or debit cards, digital wallets, or cash on delivery (COD) where available.

This innovative approach effectively addresses the challenge of navigating through urban traffic and time constraints by leveraging hyperlocal delivery networks and strategically located dark stores. It ensures that customers receive their groceries quickly, without the hassle of traffic, parking, or long store visits, making daily shopping seamless and stress-free. These approaches make grocery shopping faster, more convenient, and sustainable for both customers and the environment.

Zepto app in Improved customer satisfaction and reduced delivery time, leading to faster repeat purchase.

In conclusion, Zepto has changed the way people shop for groceries by making deliveries superfast within 10 minutes. With its real-time order tracking, customers can always know where their order is and when it will arrive. This helps solve common problems like waiting for deliveries or needing things at the last minute. By using technology and smart delivery strategies, Zepto makes grocery shopping much easier and faster for busy people. It has built trust with its customers by providing quick, reliable service and setting a new standard for fast delivery in the grocery industry.

Case Study: Real-World Problems and Zepto Innovative Solutions :-

Zepto is an Indian quick-commerce startup that aims to provide grocery deliveries within 10 minutes. It leverages a network of dark stores, cutting-edge technology, and real-time order tracking to address the growing demand for instant gratification in urban areas.

1. Problem: Urban Congestion and Traffic Challenges

Real-World Challenge: In crowded cities, heavy traffic, long travel times, and parking problems make it hard for people to shop.

Zepto's Solution: Fast Delivery Nearby

Zepto solves this problem with super-fast, 10–15-minute delivery. They use small "dark stores" (mini warehouses) near crowded areas. These stores keep groceries close to customers, so delivery is quick and skips all the traffic and parking hassles.

2. Problem: Unpredictable Demand and Inventory Management

Real-World Challenge: Traditional grocery stores often have trouble keeping the right amount of stock. They may overstock, leading to waste, or run out of popular items, especially in busy city areas where demand is unpredictable.

Zepto's Solution: AI for Smarter Stocking

Zepto uses AI and data analysis to predict what customers will buy. This helps them keep the right number of products in stock and adjust to real-time trends. By understanding customer habits, Zepto reduces waste and ensures popular items are always available.

3. Problem: Time-Consuming Shopping Experience

Real-World Challenge: People in urban areas are often pressed for time due to busy work schedules, family commitments, and other priorities. Grocery shopping, especially in crowded stores, takes valuable time that many people can't afford to spare.

Zepto's Solution: Fast and Easy Shopping

Zepto offers a 10–15 minutes delivery promise, allowing customers to get their groceries quickly and efficiently. The app simple to use, user-friendly interface, and the fast delivery ensures that customers can get what they need within minutes.

4. Problem: No Flexibility for Delivery Workers

Real-World Challenge: Gig workers often struggle with fixed schedules and irregular hours, making it hard to balance work and personal life.

Zepto's Solution: Flexible Work for Delivery Agents

Zepto offers delivery agents the freedom to choose their working hours. Agents get incentives for doing deliveries efficiently, and Zepto smart technology matches them with nearby customers and optimized routes.

Conclusion: -

Zepto solves many problems with grocery shopping in cities, like traffic, time, and high delivery costs. It uses smart technology and nearby stores to deliver orders in just 10–15 minutes. Zepto also helps delivery workers with flexible schedules. This makes shopping fast, easy, and affordable for customers while supporting workers.

Top Features of Zepto Shopping:-

1. User-Friendly App

- Available on Website, Android and Ios. advanced search and filtering for finding products quickly. AI suggests products based on purchase history and preference

2. Ultra-Fast Delivery (10–15 Minutes)

- Zepto guarantees rapid delivery of groceries within 10–15 minutes, making it perfect for last-minute needs or busy schedules.

3. Hyperlocal Dark Stores

- Zepto uses local warehouses (dark stores) to ensure products are close to customers, enabling faster delivery and reducing the need for long-distance transportation.

4. Wide Range of Products

- The product categories Includes fresh produce, dairy, beverages, packaged foods, personal care, and household essentials. user can see live updates of product availability to avoid disappointment.

5. Flexible Payment Options

- Supports credit/debit cards, UPI, digital wallets, and cash on delivery.

6. Real-Time Order Tracking

- Customers can track their order status in real time, from order placement to doorstep delivery.

7. Reliable Customer Support

- 24/7 customer support for addressing queries and resolving issues. customer can share feedback directly through the app for continuous service improvement.

8. Promotions and Discounts

- Frequent discounts and promotional offers make products more affordable. reward points or cashbacks for repeat customers.

9. Seamless Returns and Refunds

- Simple process for returning damaged or incorrect items. fast refunds credited directly to the original payment method.

Schema Description: -

The schema for Zepto involves multiple entities that represent different aspects of the platform. These entities include users, product, categories, order, order_item, payment, deliveries, reviews and dark store(warehouse), Each entity has specific attributes that describe its properties and relationship with other entities.

Users Entity

User are at the core Zepto. The user entity contains information about each user:

- **user_id (Primary Key):** A unique identifier for each user.
- **user_name:** User's display name.
- **email:** User's email address for contact and login.
- **address:** User's delivery address.
- **phone_number:** User's contact number.
- **created_at:** Timestamp when the user account was created.
- **updated_at:** Timestamp of last update to user profile.

Categories Entity

Categories entity stores information about all category.

- **category_id (Primary Key):** Unique identifier for the category.
- **name:** Name of the category.
- **description:** Optional description of the category.

Product Entity

Product entity stores information about products all available for sale.

- **product_id (Primary Key):** Unique identifier for each product.
- **name:** Name of the product.
- **image_url:** URL for the product's image.
- **description:** Detailed description of the product.
- **price:** Price of the product.
- **stock_quantity:** Available stock for the product.
- **category_id (Foreign Key referencing categories entity):** The find products in specific category.
- **created_at:** Timestamp of product creation.
- **updated_at:** Timestamp of last update to product details

Orders Entity

Orders entity stores details of user orders.

- **order_id (Primary Key):** Unique identifier for each order.
- **user_id (Foreign Key referencing users entity):** stores details of user orders.
- **order_date:** Date and time when the order was placed.
- **total_amount:** Total amount for the order.
- **status:** Order status (Pending, In Progress, Delivered, Canceled).
- **delivery_address:** Address where the order is to be delivered.
- **delivery_time_slot:** Specific delivery time or slot chosen by the user.
- **payment_status:** Status of the payment (Paid, Pending).
- **delivery_method:** Type of delivery (Standard, Express).
- **created_at:** Timestamp of order creation.
- **updated_at:** Timestamp of last update to order details.

Order Items Entity

Order Item entity stores information all total order Item.

- **order_item_id (Primary Key):** Unique identifier for each order item.
- **order_id (Foreign key referencing orders entity):** The order item to a specific order.
- **product_id (Foreign key referencing products entity):** The order item to a specific product.
- **quantity:** Number of units of the product in the order.
- **unit_price:** Price of the product at the time of the order.
- **total_price:** Total price for the order item (quantity * unit price).

Payments Entity

Payment's entity stores payment information for user.

- **payment_id (Primary Key):** Unique identifier for each payment transaction.

- **order_id (Foreign key referencing orders entity):** The payment to a specific order.
- **payment_date:** Date and time when the payment was made.
- **amount_paid:** Total amount paid by the customer.
- **payment_method:** Method of payment
- **payment_status:** Status of the payment
- **transaction_id:** Unique transaction ID provided by the payment gateway.
- **created_at:** Timestamp of payment creation.

Deliveries Entity

Deliveries entity stores delivery person and status information.

- **delivery_id (Primary Key):** Unique identifier for each delivery.
- **order_id (Foreign Key referencing orders entity):** The Delivery to a specific order.
- **delivery_person_name:** Identifier for the delivery person name.
- **delivery_address:** Delivery address.
- **delivery_time:** Timestamp when the delivery was made.
- **status:** Delivery status (Pending, In Transit, Delivered).
- **delivery_notes:** Additional information about the delivery.
- **created_at:** Timestamp of order delivery creation.
- **updated_at:** Timestamp of last update to order details.

Reviews Entity

Reviews entity stores product review information for user completed orders.

- **review_id (Primary Key):** Unique identifier for each review.
- **product_id (Foreign key referencing product entity):** The review to a specific product.
- **user_id (Foreign Key referencing users' entity):** The user who submitted the review.
- **rating:** Rating score (1 to 5 stars).
- **review_text:** Customer-written review.
- **created_at:** Timestamp when the review was submitted.

9. Dark Stores (Warehouse) Entity

Dark Stores entity stores details about warehouse location.

- **store_id (Primary Key):** Unique identifier for each dark store.
- **product_id (Foreign key referencing product entity):** The review to a specific product.
- **location:** Geographical location of the store.

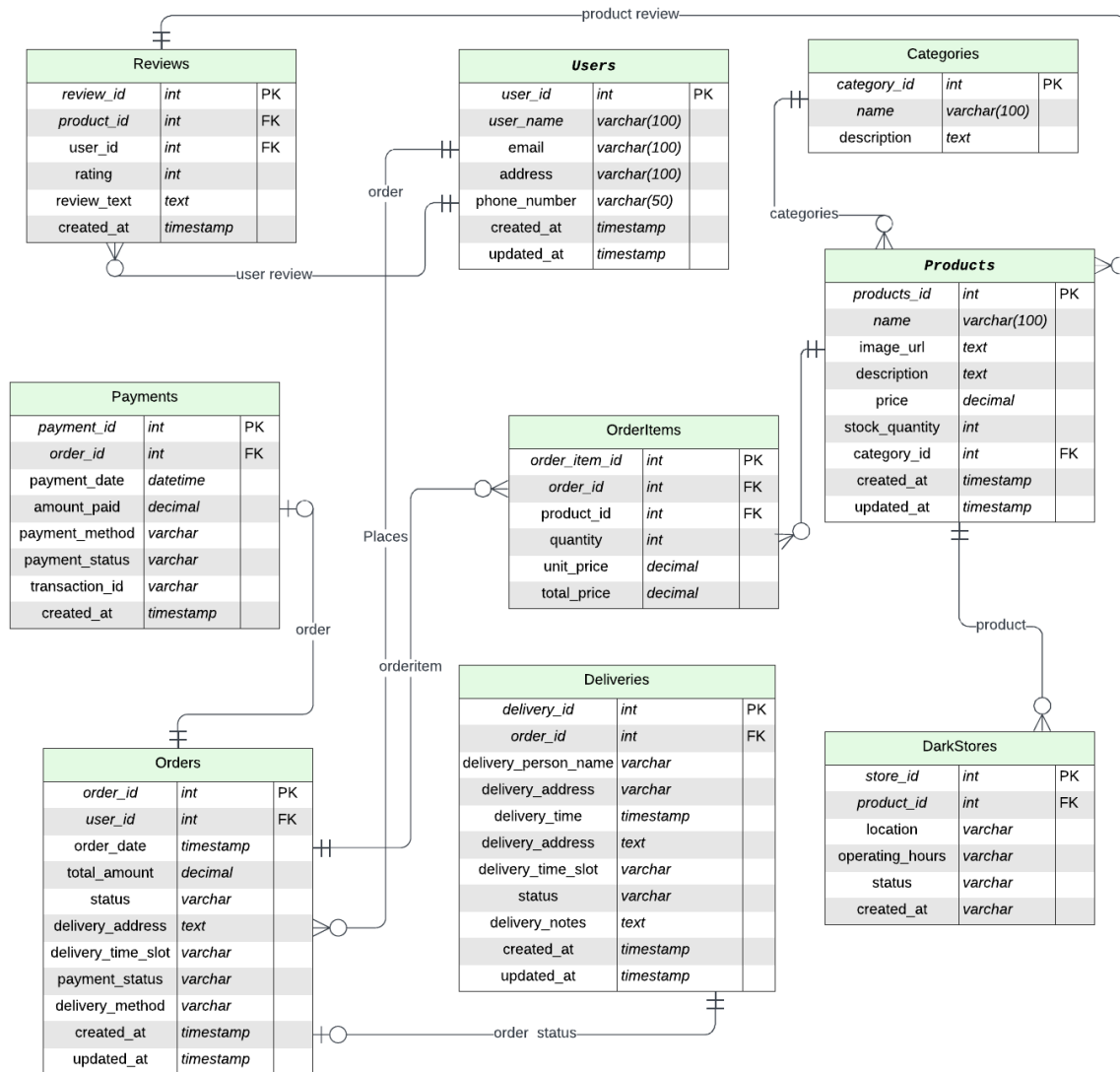
- **operating_hours:** Store's operating hours.
- **status:** Whether the store is operational or not.
- **created_at:** Timestamp when the store was created.

Relationships Entity: -

- Each User can have many **Orders** (One-to-many relationship).
- Each Order can have many **Order Items** (One-to-many relationship).
- Each Product can have many **Order Items** (One-to-many relationship).
- Each Order can have one **Payments** (One-to-one relationship).
- Each Order can have one **Deliveries** (One-to-one relationship).
- Each Product can have one **Reviews** (One-to-many relationship).
- Each Users can have one **Reviews** (One-to-many relationship).
- Each Dark Stores can have one **Products** (Many-to-many relationship).

ER Diagram:

Let's construct an ER diagram that vividly portrays the relationships and attributes of the entities within the Zepto schema. This ER diagram will serve as a visual representation, shedding light on the pivotal components of Zepto data model. By employing this diagram, you'll gain a clearer grasp of the intricate interactions and connections that define the platform's dynamics.



Conclusion

In this case study, the schema diagram for the Zepto Grocery Delivery app includes several key entities such as User, Category, Product, Order, Order Item, Payment, Deliveries, Dark store and Review, each with its set of attributes and relationships. These entities work together to model the core functionality of the application, allowing customers to shop for products, place orders, make payments, manage their deliveries carts, and track shipments. The diagram provides a simplified representation of the database structure, and in practice, a real-world database for a platform as comprehensive as Zepto would be more intricate, incorporating additional tables and attributes to support its extensive features and services.